

Jakob Olof Popov Wirén

(+46)76286389 , jakowiren@gmail.com

2000-04-06, Swedish Citizen

EDUCATION

Lund University, LTH

Lund, Sweden

M.Sc. 5, Engineering Physics

Aug 2019 – Jan 2025

- Master's specialization in *Control Theory*
- Master thesis: **Machine Learning for Anti-Poaching: Decision Tree Applications on the Savannah**. Worked in Limpopo, South Africa to develop three anti-poaching systems with decision trees.
- Decision trees, MLFlow, Bayesian Optimization, data collection and creative thinking

Technical University Berlin

Berlin, Germany

Erasmus Exchange

Oct 2022 - Aug 2023

- Relevant coursework: **Project – Computer Vision for Remote Sensing, State Estimation for Robotics, Event-Based Robot Vision, Machine Learning 1+2, Applications of Robotics and Autonomous Systems**
- Master's GPA: 1.88 (German Grading System)

ESA Academy

ESEC-Galaxia, Belgium

Selected Student

- *ESA/ELGRA Gravity-Related Research Summer School*
- *Ladybird Guide to Spacecraft Communications Training*

Jun 2021

Mar 2023

Kungsholmens Gymnasium

Stockholm, Sweden

Natural Science

Aug 2016 – Jun 2019

- Overall GPA: 22.50/22.50

PROFESSIONAL EXPERIENCE

XER Technologies

Zurich, Switzerland

Machine Learning Intern

Mar 2025 – Dec 2025

- A small company. I pushed for creating an organized and maintainable software and data infrastructure. Built tools for internal and eventually customer use, including data visualization, fleet management and real-time flight displays in Unity.
- AWS, Docker, Python, Unity, C#, JS, React, Nginx.

Volvo Cars R&D

Lund, Sweden

Software Intern

Jul 2024 – August 2024

- Developed unit tests in C++ and Robot tests. Manually flashed and tested new release software on test vehicles. Worked within the Connectivity Team. Skills developed in C++.

Bosch R&D

Lund, Sweden

Software Intern

Jun 2022 – Sep 2022

- Created an automated testing platform for Bluetooth in Android-based In-Vehicle-Infotainment systems. Skills developed in Python, Robot Framework, Appium and RPI4.

EXTRACURRICULAR ACTIVITIES

Lund Formula Student

Lund, Sweden

Manager of the Driverless Subteam

Jun 2021 – Sep 2022

- Leading and managing the Driverless Subteam with a big-picture mindset - developing, testing and competing with an autonomous F1 car. Giving advice and working in ROS1, python and C++ to help the subteam develop code for perception, sensor-fusion, SLAM and path-planning. Participated in international engineering competitions in Hungary and Czech Republic.

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CERN

HSSIP Intern

Geneva, Switzerland

Jun 2018

- Preparation work of spectrometers for an upcoming experimental campaign. Doppler-shift correction, manual manipulation and calibration of angular position(s) of radiation detectors, cross-referencing of practical data and theoretical data in order to determine sources of background radiation.

Nanyang Technological University

NTU, Singapore

Voluntary research assistant

Jul 2017

- One-month voluntary research work spent conducting simple experiments, assembling equipment and doing secondary research for Assistant Professor Tsao Hoi Nok with his research regarding "An entirely light driven optical memory for energy efficient electronics".

F-Guild

Lund, Sweden

President of Equality Committee

Jan 2021 - Jan 2022

- Acted as student safety representative and led a team of six volunteers working on issues in psychosocial health and equality at the F-guild. Organized larger-scale events together with the LTH equality board, including arranging for guest-speakers with professional background in the area. The post entailed professional secrecy.

CONFERENCES

Ocean Park International STEAM Education Conference

Hong Kong, China

Poster presenter

Jun 2019

Poster titled "Engaging peers, teachers and the school board in local climate activism: The experience of creating "KG Earth" association for spreading of environmental awareness".

OTHER ACTIVITIES

Sweden Rock

Sölvesborg, Sweden

Bar Staff

Jun 2022, Jun 2023, Jun 2024

Volunteering around the world

- Esperanza Verde animal rescue center, Amazonas Peru Jun-Aug 2018
- Indómito hostel, Bariloche Argentina Oct 2023
- Yambolombia hostel, Salento Colombia Feb 2024
- SanWild sanctuary, Limpopo South Africa Sep-Dec 2024

Hackathons

Andon Labs Hackathon - 1st place

Jan 2025

SKILLS AND INTERESTS

- **Skills:** Problem solving, Working in a team
- **Languages:** Swedish – C2, English – C2, Spanish – C1, Russian – A1
- **Interests:** Intercultural experiences, Programming, Conservation, Drinking coffee